A REVIEW AID FOR GERMAN CREDIT APPLICANTS

Credit Risk Auditor

Score, explanation, lending clauses, human review

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CONTEXT

The problem

Flag vs. Reason

Model flags lack actionable justification.

Weight vs. Policy

Positive weights may violate adverse rules.

Label vs. Score

Historical outcomes differ from risk scores.

ARCHITECTURE

What the system does

1

XGBoost

Scores predicted default risk.

2

SHAP

Maps path from average risk.

3

Clauses

Retrieves official EBA guidelines.

4

Draft

Writes optional reviewer summary.

5

Human

Makes final lending decision.

DATASET

Data

1,000

Applicants, 20 Attributes

UCI / Statlog German Credit

Binary classes: Class 1 good, class 2 bad credit.

Target encoding: Bad credit modeled as positive class.

Checking status: Balance below 0 DM.

CLASSIFIER

Model

78%
Held-out accuracy

0.815
ROC-AUC score

50%
Reviewer flag cutoff

160 KB
Shared artifact size

Architecture: XGBoost, depth 4, 80/20 split, seed 42.

Reproducibility: Identical scores across all machines.

AUDIT GROUP: UNDER 25

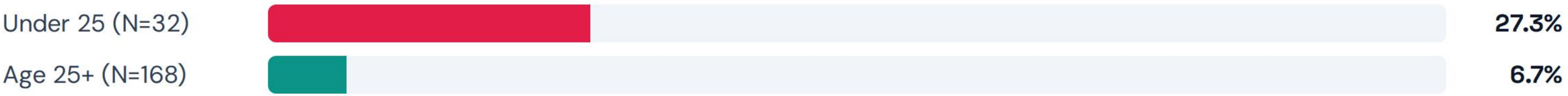
Fairness

Favorable Rate

Gap: -15.3 pts (-9.4 pts without age)



False-Positive Rate



Sex is coupled with marital status; non-foreign test group has only 9 applicants.

RETRIEVAL & GENERATION

Clauses and the draft

Policy Corpus

Project paraphrases: Eight EBA/GL/2020/06 clauses indexed once.

Official guideline: Full 89-page report grounds search.

Draft Guardrails

Citations & review: Flags unretrieved IDs or missing “manual review.”

Prohibited claims: Warns on loan denial, “grounds for denial,” or invented missed payments.

Completeness: Flags any sentence that stops halfway.

Live demo

INTERPRETABILITY

How to read the explanation

Baseline: Starts at 28% average default risk.

Red bars: Push predicted risk upward.

Teal bars: Lower predicted risk downward.

Dual views: Top bar shows percent; log-odds show additive math.

Observed risk indicators

Only two observed indicators qualify: checking below 0 DM and past payment delays.

A positive SHAP value is not itself an adverse reason.

GOVERNANCE

Limits and choices

100% Human Referral

No calibrated auto-decision band or rejection cutoff.

Small Sample Size

1,000 rows total; 32 under-25 test applicants.

Draft Hallucination Risk

Automated warnings flag invented facts for reviewers.

Deterministic Output

Fixed model file guarantees identical SHAP explanations.

Thank you for your attention.
Questions?